

Sayyed Mazhar

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Summary

Final-year Computer Engineering student with hands-on experience in **AI/ML system development**, full-stack web deployment, and data analytics. Built and deployed Sentinel AI — a real-time violence detection system using a **hybrid ConvNeXt-Tiny + LSTM + Temporal Attention architecture** achieving 96.5% accuracy — featuring a live React SOC dashboard, FastAPI backend, WebSocket alerts, and full cloud deployment (Vercel + ngrok). Experienced in Python, PyTorch, FastAPI, React, SQL, and Power BI. Proven ability to take ML projects from research to production.

Education

Bachelor of Technology in Computer Science (B.Tech), *K.C.E College Of Engineering* 11/2022 – Present

Technical Skills

AI / ML

PyTorch, ConvNeXt, LSTM, Temporal Attention, Transfer Learning, Deep Learning, Computer Vision, Video Classification

Backend

FastAPI, Uvicorn, WebSockets, REST API, Python, CORS, Lifespan Events, Email Alerts (SMTP)

Frontend

React 18, Tailwind CSS, Recharts, Leaflet.js, WebSocket API, Lucide React, Hooks (useState, useEffect, useMemo, useRef)

Data Analysis & Visualization

Pandas, NumPy, SQL, Excel, Exploratory Data Analysis. Power BI (DAX, Interactive Dashboards), Matplotlib, Recharts, Excel Charts.

DevOps & Deploy

Git, GitHub, Vercel, Render, ngrok, Hugging Face Hub, Virtual Environments, CI/CD

Practical Experience

Data Analytics internship, *Edu Tantr pvt Ltd*

11/2025 – 02/2026
Bangalore

- Completed a Data Analytics internship, gain experience in data collection, cleaning, visualization, and analysis using industry-standard tools.
- Built an Airline Management System project, analyzing operational data to uncover insights on flight performance and passenger trends through end-to-end data pipeline development.
- Demonstrated strong analytical thinking by translating raw data into actionable business insights, supporting data-driven decision-making throughout the internship.

Python Development-Based, *Paarsh Infotech Pvt Ltd*

2025

- Gained hands-on experience in Python programming by building logic-driven programs and mini projects.
- Worked with loops, functions, conditionals, file handling, and data structures (lists, dictionaries, sets).

Projects

Sentinel AI — Real-Time Violence Detection System,

11/2025 – 03/2026

(PyTorch, FastAPI, React, WebSocket, Vercel, ngrok)

- Designed and deployed a full-stack AI violence detection system using a hybrid ConvNeXt-Tiny + LSTM + Temporal Attention architecture achieving 96.5% test accuracy and 0.9805 ROC AUC on benchmark datasets
- Built FastAPI backend with 5 endpoints (/predict, /health, /history, /ws/alerts) includes file validation, 3- zone confidence threshold system (Violent/Uncertain/Non-Violent), structured logging, and automated Gmail SMTP email alerts
- Developed a professional React SOC dashboard with 5 tabs: live camera simulation, interactive Jalgaon campus map (Leaflet.js + OpenStreetMap), forensic video analysis with temporal attention heatmap, ML performance charts (Recharts), and configuration panel
- Deployed frontend on Vercel (auto CI/CD from GitHub) and backend via ngrok tunnel; model (345MB) hosted on Hugging Face Hub with automatic download on startup
- GitHub: github.com/Mazhar2905/sentinel-ai | Live: sentinel-ai-eosin.vercel.app

Road Accident Analysis Dashboard, *(Python, Pandas, NumPy, Power BI, Excel)*

11/2025

- Analyzed 50,000+ road accident records across Indian states to identify high-risk regions, year-wise trends, and accident causes
- Built interactive Power BI dashboard with state-level comparisons, time-series analysis, and cause breakdown visuals supporting policy planning insights.
- Applied data cleaning and EDA to identify outliers and seasonal patterns in accident frequency

Electronics Sales Dashboard, *(SQL, Excel, Power BI)*

09/2025

- Designed Power BI dashboard analyzing sales performance across laptops, mobiles, tablets, and accessories with KPIs: Total Sales, Profit, AOV, Quantity Sold
- Used SQL for data extraction with joins and aggregate functions; implemented DAX measures for dynamic quarter and product-type filters
- Identified top-performing products and seasonal trends enabling data-driven inventory and pricing decisions

Local Business Lead Generation Pipeline, *(Python, Selenium, OpenStreetMap API, Google Sheets)*

- Built automated lead scraping pipeline using Selenium (JustDial) and OpenStreetMap Overpass API to collect local Jalgaon business contacts
- Logged structured lead data (name, phone, category, location) to Google Sheets via API for sales outreach pipeline

Relevant Coursework / Certifications

SQL Development, *Simplilearn***Ultimate Data Science course**, *codewithharry***Python Training Completion**, *EduPyramids*